# Optimization IV
# Technical Issues in Optimization

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June 5, 2025

# Content

- Constraint qualification conditions (NDCQ)
- Envelope Theorem
- Review

# NDCQ
# Non-Degenerate Constraint Qualification

# CQ in the 2-variable case

Recall the 2-variable situation.

$$\max z = f(x, y)$$
$$s.t.\ g(x, y) = c$$

The optimal point $(x^*, y^*)$ is the point where the slopes of $f$ and $g$ are equal. We constructed the expression

$$-\frac{f_x}{f_y} = -\frac{g_x}{g_y} \iff \frac{f_x}{g_x} = \frac{f_y}{g_y} = \lambda$$

- Note 1: The above expression is only valid when $g_x, g_y \neq 0$ at $(x^*, y^*)$. Therefore, we must ensure this condition on the constraint holds when using the Lagrangian.
- Note 2: If the constraint is linear, this condition is automatically satisfied.
- Note 3: We must assume that $g_x$ or $g_y$ or both are nonzero at the optimal point.

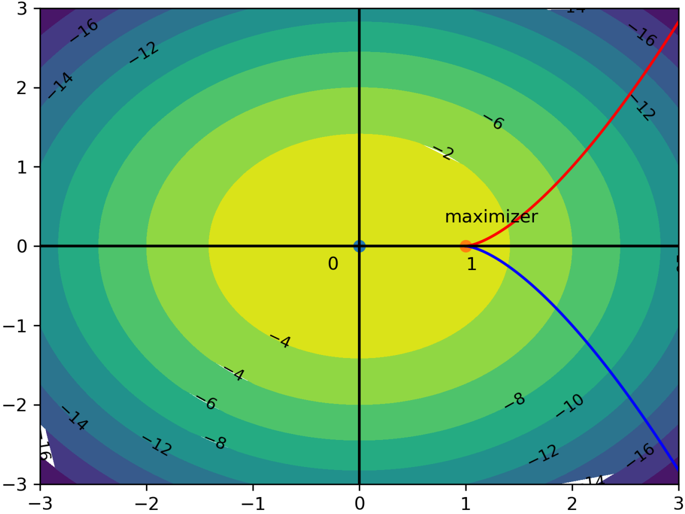
# When Lagrangian Fails

Example:

$$\max f(x,y) = -x^2 - y^2$$
$$s.t. g(x,y) = (x-1)^3 - y^2 = 0$$

- Determine the domain $D(x, y)$.
- Use the Lagrangian method. Do you find a solution?
- Use the substitution method. Do you find a solution?
- Continue the substitution method, show that $f'(x) < 0$ and find the optimal point.

# On the Model



In this case, we can see that:

- The Lagrangian method does not determine the optimal point.
- At the optimal point $(x_0, y_0) = (1, 0)$, the gradient of the constraint is

$$Dg = \nabla g^T(x_0, y_0) = \begin{pmatrix} 2(x_0 - 1)^2 & -y_0^2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}$$

  In other words, the gradient at the critical point is degenerate.
- The Lagrangian is only applicable when the constraint function is continuously differentiable and the gradient of the constraint at the critical point is nonzero (non-degenerate).

For 2 variables, 1 constraint, the CQ is

$$Dh(x_1, x_2) = \begin{pmatrix} h_{x_1}(x^*) & h_{x_2}(x^*) \end{pmatrix} \neq \begin{pmatrix} 0 & \dots & 0 \end{pmatrix}$$

For $n$ variables, 1 constraint, the CQ is

$$Dh(x_1, \dots, x_n) = \begin{pmatrix} h_{x_1}(x^*) & \dots & h_{x_n}(x^*) \end{pmatrix} \neq \begin{pmatrix} 0 & \dots & 0 \end{pmatrix}$$

For $n$ variables, $m$ constraints,

$$rank\ \ Dh(x^*) = \begin{pmatrix} h^1_{x_1} & h^1_{x_2} & \dots & h^1_{x_n} \\ h^2_{x_1} & h^2_{x_2} & \dots & h^2_{x_1} \\ \vdots & \vdots & \ddots & \vdots \\ h^m_{x_1} & h^m_{x_2} & \dots & h^m_{x_n} \end{pmatrix} = m$$

The rank of the Jacobian matrix of $m$ constraints is $m$. Then we say the problem satisfies the Non-Degenerate Constraint Qualification (NDCQ).

1. $$\max f(x,y) = xy$$ $$s.t. g(x,y) = x + y = 16$$

   Satisfies CQ because $\nabla g = \begin{pmatrix} g_x & g_y \end{pmatrix} = \begin{pmatrix} 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 \end{pmatrix}$ $\forall (x,y)$.

2. $$\max f(x,y) = x^2 y$$ $$s.t. g(x,y) = 2x^2 + y^2 = 3$$

   Satisfies CQ. We have $\nabla g = \begin{pmatrix} g_x & g_y \end{pmatrix} = \begin{pmatrix} 4x & 2y \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}$ if and only if $(x,y) = (x_0, y_0) = (0,0)$ but $2x_0^2 + y_0^2 \neq 3$ (the degenerate point does not lie on the constraint).

3. $$\max f(x,y) = -x^2 - y^2,$$ $$s.t. g(x,y) = x - y^3 = 0$$

   Satisfies CQ. We have $\nabla g = \begin{pmatrix} g_x & g_y \end{pmatrix} = \begin{pmatrix} 1 & -2y^3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}$ if and only if $1 = 0$ and $y = 0$ (impossible).

# Envelope Theorem

Purpose: To see how the optimal value of the objective function changes when parameters change.

## Envelope Theorem for Unconstrained Optimization

Let $f(x, a)$ be differentiable with $x \in \mathbf{R}^n$ and $a$ a parameter. For each parameter $a$, consider the optimization problem $\max_x f(x, a)$. Let $x^*(a)$ be the solution, and further assume $x^*$ is differentiable with respect to $a$, then:

$$\frac{df}{da}(x^*(a), a) = \frac{\partial f}{\partial a}(x^*(a), a)$$

Proof

# Example

When $a$ increases, how does the optimal value $f(x^*, a)$ change, given the objective function

$$\max f(x, a) = -x^2 + 2ax + 4a^2 \tag{1}$$

FOC:

$$f'(x) = -2x + 2a = 0$$

giving the optimal point $x^*(a) = a$.

- Method 1: substitute into the original objective function

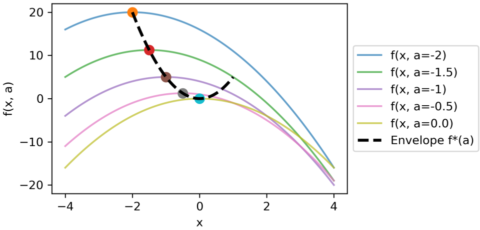
$$f(x^*(a), a) = f(a) = 5a^2 \tag{2}$$

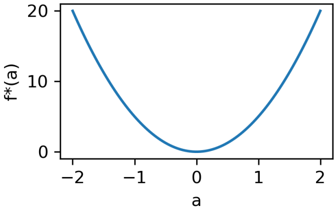
  giving $f^{*\prime}(a) = 10a$
- Method 2: use the envelope theorem, differentiate directly from (1)

$$f^{*\prime}(a) = 2x^* + 8a = 10a$$

  Thus, we can skip step (2).

# Illustration



## Envelope Theorem for Constrained Optimization

Let $f(x, a)$ be differentiable with $x \in \mathbf{R}^n$ subject to constraint $h(x, a) = 0$, and $a$ a parameter. For each parameter $a$, consider the optimization problem $\max_x f(x, a)$ s.t. $h(x, a) = 0$. Let $x^*(a)$ be the solution. Assume $x^*$ is differentiable with respect to $a$, and the constraint satisfies NDCQ, then:

$$\frac{df}{da}(x^*(a), a) = \frac{\partial L}{\partial a}(x^*(a), \lambda(a), a)$$

where $L$ is the Lagrangian and $\lambda(a)$ is the corresponding Lagrange multiplier.

Proof

In the cost minimization problem

$$\min C(K, L) = rK + wL$$

s.t.

$$F(K, L) = Q \ (0 < \alpha < 1)$$

Let $C^* = C^*(r, w, Q)$. Compute $\partial C^*/\partial r$, $\partial C^*/\partial w$, $\partial C^*/\partial Q$

Lagrangian function

$$L = rK + wL - \lambda(F(K, L) - Q)$$

Using the envelope theorem

$$(r) : \frac{\partial C^*}{\partial r} = K^*,$$
$$(w) : \frac{\partial C^*}{\partial r} = L^*,$$
$$(Q) : \frac{\partial C^*}{\partial Q} = \lambda.$$

Assuming $F(K, L) = AK^\alpha L^{1-\alpha}$, one can fully compute $K^*, L^*, \lambda$.

# Review

# Price Discrimination [EMEA:Eg13.4.1]

A monopolist sells the same product in two markets with the following demand functions: (all $a_1, a_2, b_1, b_2, \alpha$ are positive and $a_1, a_2 > \alpha$)

$$p_1 = a_1 - b_1 q_1,$$
$$p_2 = a_2 - b_2 q_2$$

Total cost is

$$C(Q) = \alpha Q = \alpha(q_1 + q_2)$$

1. Write the profit function.
2. Find $(q_1^*, q_2^*)$ to maximize profit.
3. What prices will the monopolist set in each market?
4. Find the monopoly profit.
5. What if the government prohibits price discrimination? Compute the loss, given $a_1 = 100, a_2 = 80, b_1 = b_2 = 1, \alpha = 20$.

# Representative Agent Problem

A representative agent consumes good $x$ and chooses working time $\ell$. The utility function is as follows:

$$U(x,\ell) = \alpha \ln x + \beta \ln(1-\ell) \tag{3}$$

s.t.

$$px = w\ell + b$$

given that $0 \leq \ell \leq 1$ and $x, w, \alpha, \beta > 0$. Where $p$ is the price of goods, $w$ is wage, $b$ is other income.

1. Find $(x^*, \ell^*)$ to maximize utility. Check SOC.
2. How does an increase in wage $w$ affect optimal decisions?
3. Prove that $x^*(\mu p, \mu w, \mu b) = \mu^0 x^*(p, w, b) \forall \mu > 0$.
4. If $b$ increases to $(1+\gamma)b$ with $\gamma$ being a small positive number, what happens to the new optimal utility? Compare the Lagrange multiplier with $\partial U/\partial b$.

# Women's Opportunity Cost Problem

A family consists of a wife and a husband. The responsibility of childbearing $n$ and educating children $e$ falls on the woman. The husband devotes all his time to work. The woman's time constraint is

$$l_f + n\phi = 1$$

where $l_f$ is the remaining time the woman spends working. With $w_m, w_f$ being wages for men and women, this family has the utility function:

$$u(c, n, e) = \ln c + \delta \ln n + \delta\gamma \ln(\theta + e)$$

s.t.

$$c + en \leq w_m + l_f w_f,$$
$$e \geq 0.$$

1. Find $c^*, n^*, e^*$ to maximize utility for the family. Under what condition on $w_f$ is $e^* = 0$? Compare $n^*$ when $e^* = 0$ and when $e^* > 0$.
2. What happens to $n^*$ if $w_m/w_f$ narrows?

# Appendix

# Envelope Theorem for Unconstrained Optimization

Take the total derivative of both sides

$$\begin{aligned}\frac{df}{da}(x^*(a),a) &= \frac{\partial f}{\partial x_1}(x^*(a),a)\frac{dx_1^*}{da} + \cdots + \frac{\partial f}{\partial x_n}(x^*(a),a)\frac{dx_n^*}{da} \\ &\quad + \frac{\partial f}{\partial a}(x^*(a),a)\frac{da}{da} \\ &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(x^*(a),a)\frac{dx_i^*}{da} + \frac{\partial f}{\partial a}(x^*(a),a)\end{aligned}$$

Since $\frac{\partial f}{\partial x_i}(x^*(a),a) = 0$ for $i = 1, \ldots, n$ by FOC, the above equation reduces to

$$\frac{df}{da}(x^*(a),a) = \frac{\partial f}{\partial a}(x^*(a),a)$$

# Envelope Theorem for Constrained Optimization

Let $V(a) = f(x^*(a), a)$, we take the total derivative

$$\frac{dV}{da} = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\frac{dx_i^*}{da} + \frac{\partial f}{\partial a}\frac{da}{da} = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\frac{dx_i^*}{da} + \frac{\partial f}{\partial a} \tag{4}$$

Lagrangian function

$$L = f(x(a), a) - \lambda(a) g(x(a), a)$$

According to the FOC, we have for $i = 1, \ldots, n$

$$\frac{\partial L}{\partial x_i}(x^*(a), a) = \frac{\partial f}{\partial x_i}(x_i^*(a), a) - \lambda(a)\frac{\partial g}{\partial x_i}(x^*(a), a) = 0$$

i.e.,

$$\frac{\partial f}{\partial x_i}(x^*(a), a) = \lambda(a)\frac{\partial g}{\partial x_i}(x^*(a), a)$$

Thus, the right-hand side of (4) can be rewritten as

$$\frac{dV}{da} = \lambda(a) \sum_{i=1}^{n} \frac{\partial g}{\partial x_i}(x^*(a), a) + \frac{\partial f}{\partial a} \tag{5}$$

Remember that the constraint is always satisfied: $g(x(a), a) = 0$. Take the total derivative of both sides

$$\frac{dg}{da} = \sum_{i=1}^{n} \frac{\partial g}{\partial x_i} \frac{dx}{da} + \frac{\partial g}{\partial a} \frac{da}{da} = 0$$

From which we obtain

$$\sum_{i=1}^{n} \frac{\partial g}{\partial x_i} \frac{dx}{da} = -\frac{\partial g}{\partial a} \tag{6}$$

Substituting (6) into (5) gives

$$\frac{dV}{da} = \frac{\partial f}{\partial a} - \lambda(a) \frac{\partial g}{\partial a} \equiv \frac{\partial L}{\partial a}$$